

terminology binding with SNOMED CT

DATE

21 September 2026

TIME

13:00–17:00 CEST

FORMAT

Online

LANGUAGE

English

A four-hour technical session for national implementation stakeholders and eHealth software developers.

SNOMED INTERNATIONAL

workshop schedule

PART 1 · FOUNDATIONS, ENVIRONMENT AND EXPRESSION CONSTRAINTS

13:00 - 13:10 Welcome and introduction

13:10 - 13:45 SNOMED CT fundamentals

13:45 - 14:25 Terminology server setup

14:25 - 15:05 Expression Constraint Language

15:05 - 15:20 BREAK ☕

PART 2 · TERMINOLOGY BINDINGS AND PRACTICAL EXERCISES

15:20 - 15:55 Terminology bindings

15:55 - 16:30 Terminology Binding Demonstrator

16:30 - 16:50 FHIR ConceptMaps in practice

16:50 - 17:00 Wrap-up, questions and next steps

Note: This is a hands-on session.
Licensing, access to SNOMED CT release files and Docker
should already be in place.

PART 1 · SECTION 01

welcome and introduction

from national requirement to working implementation

THE REQUIREMENT

National programmes specify SNOMED CT

Specifications say which fields must be coded. They may not say which concepts are valid in each field, that gap is filled by a binding.

THE GAP

Any code is legal until the binding says otherwise

Without a binding, a diagnosis field can legally hold a procedure or a body structure. Data passes validation and still cannot be reused.

AFTER TODAY

You will have built the whole chain

A local terminology server, ECL-defined value sets, FHIR bindings expressed as resources, and ConceptMaps translating local codes, all running on your own machine.

by the end of the session, **you will be able to**

-
-  Explain what a terminology binding is, and how bindings connect an information model to SNOMED CT.
 -  Describe the role of a terminology server in supporting bindings, and run a local instance.
 -  Use Expression Constraint Language (ECL) to define the value sets that bindings refer to.
 -  Express and maintain bindings using FHIR terminology resources and operations.
 -  Apply these techniques in practice through guided, hands-on exercises.
-

interactive learning



Theory with exercises

Each part pairs a short explanation with practical work, server setup, ECL, bindings and maps on your own machine.



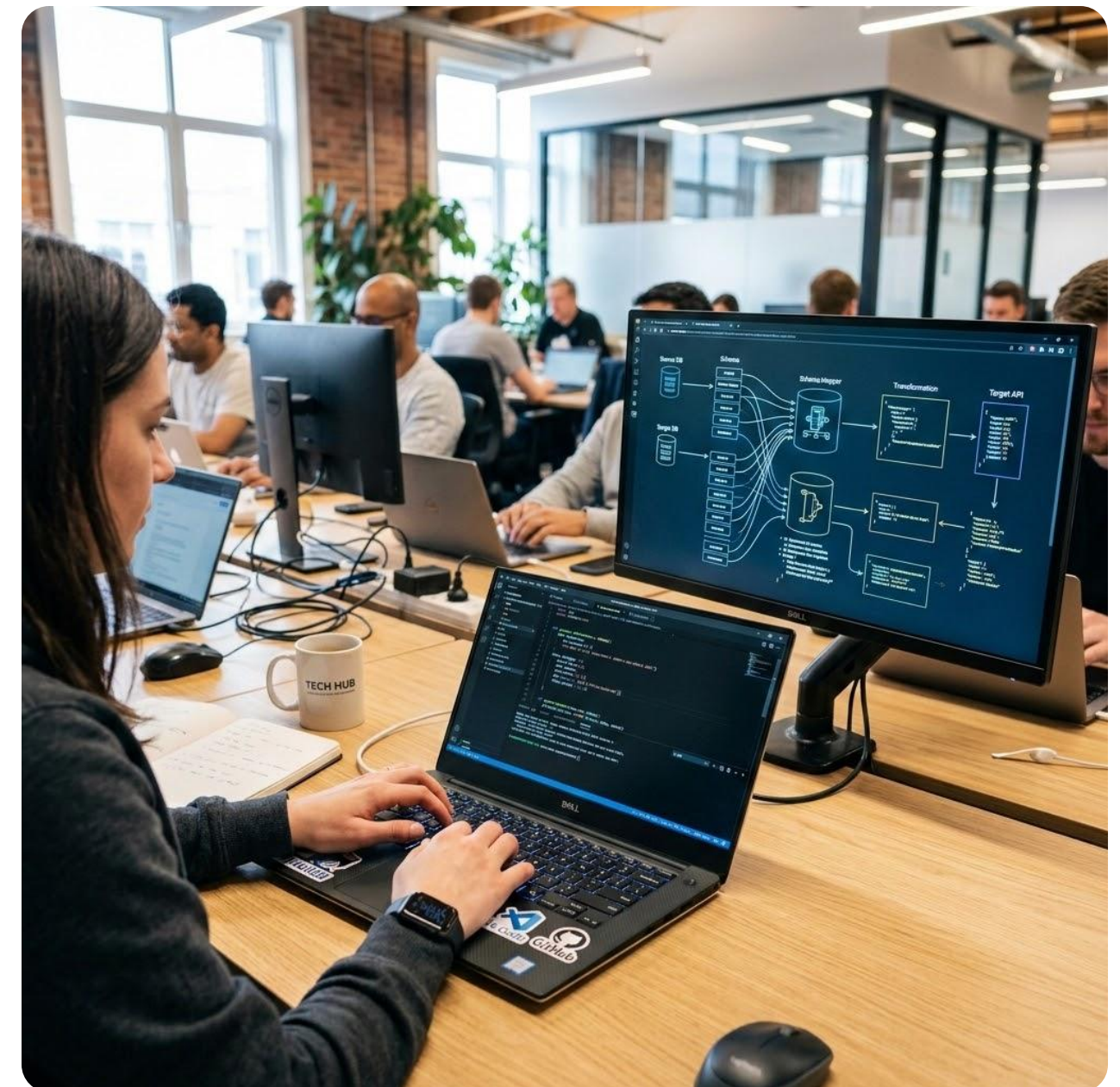
Active involvement

Your participation is key. We expect hands-on engagement throughout the four hours.



Communication

Use the chat for comments, or raise your hand for questions at any time. Two screens help: one for the call, one for the exercises.



implementation support team



Alejandro Lopez Osornio

Senior Implementation Support Specialist, SNOMED International



Anne Randorff Højen

Implementation Support Specialist, SNOMED International

OUR MISSION

To collaborate with Members to achieve their SNOMED CT implementation goals, and realise maximum value from SNOMED CT.

OVER TO YOU, IN THE CHAT

who are you?

Your role, your organisation, and your expectations for today



Join at:
[ahaslides.com/
281OE](https://ahaslides.com/281OE)

PART 1 · SECTION 02

SNOMED CT fundamentals

Concepts, descriptions and relationships; the concept model; reference sets and how editions and extensions fit together.

INTRODUCTION TO SNOMED CT

the world's most comprehensive multilingual clinical terminology.

360K+

ACTIVE CONCEPTS

1M+

DESCRIPTIONS AND SYNONYMS

53

MEMBER COUNTRIES

a graph of clinical knowledge, not a code list

Concepts

A unique clinical meaning with a stable numeric identifier (SCTID).

Descriptions

The human-readable terms for a concept, fully specified name, preferred term and synonyms, in any language.

Relationships

Typed links, *is a* for the hierarchy, attributes such as *finding site* for the concept model.

EXAMPLE · ONE CONCEPT IN CONTEXT

73211009 |Diabetes mellitus (disorder)|

is a Disorder of endocrine system

is a Disorder of glucose metabolism

finding site Endocrine system structure

interprets Glucose metabolism

each concept carries multiple human-readable descriptions

CONCEPT · ID 22298006	
Myocardial infarction	
FSN	Myocardial infarction (disorder)
PREFERRED	Myocardial infarction
SYNONYM	Cardiac infarction
SYNONYM	Heart attack
SYNONYM	Infarction of heart

Designed for both **humans** and **machines**.

The **fully specified name** disambiguates by semantic tag, (disorder), (procedure), (finding), and is unique system-wide.

Each concept also carries one **preferred term** per language, plus any number of **synonyms** , so the interface can show what the clinician types while the record stores the stable identifier.

Language reference sets decide which term is preferred or acceptable per language and dialect, the same mechanism a national translation uses.

disorders reflect **real-world complexity** through multiple categories

MULTI-PARENT CLASSIFICATION

Disorder of lung

Infectious disease

↑ IS A

.

IS A ↑

Pneumonia

Defining attributes

Finding site: Pulmonary structure

Associated morphology: Inflammation

Richer retrieval

The same concept is reachable through several clinical pathways, so a query framed either way finds it.

Decision support

Complex knowledge is represented accurately rather than forced into one category.

Consequence for bindings

A value set defined as "descendants of *Disorder of lung* " picks up pneumonia automatically, you do not enumerate it.

every concept inherits from **top-level hierarchies** through **is a**

Clinical finding (top level)

↑ IS A

Disorder of lung

↑ IS A

Pneumonia

↑ IS A

Rubella pneumonia

Inferred: Rubella pneumonia is a Clinical finding

Implicit knowledge

The hierarchy carries the chain of knowledge, so nobody has to state the obvious.

Powerful retrieval

A query for all lung disorders correctly includes every specific descendant, now, and in future releases.

Logical consistency

Description logic keeps classification sound across the whole terminology.

SNOMED CT has nineteen top-level hierarchies

 Body structure	 Clinical finding	 Environment / location	 Event	 Observable entity
 Organism	 Pharmaceutical product	 Physical force	 Physical object	 Procedure
 Qualifier value	 Record artifact	 Situation with explicit context	 Social context	 SNOMED CT model component
 Special concept	 Specimen	 Staging and scales	 Substance	

Broad coverage: Designed to meet all requirements for clinical data in an EHR.

Note: Common clinical-record contexts (signs, tests, diagnosis, medication) are captured across these hierarchies. Choosing the right one is the first decision in any binding.

flexible features for clinical implementation



Continuous releases

Monthly International Edition releases adapt to emerging clinical requirements. Robust versioning and history tracking



Extendable

National and local extensions to meet national use cases for implementation and use



Subsets

Specify a subset as an enumerated list or as an ECL query, the choice you will make in the exercises.



Maps

Declare maps between SNOMED CT and other code systems, distributed as refsets or FHIR ConceptMaps.



Computable query language

ECL expresses the rules behind a value set, and servers evaluate them on demand.

understanding SNOMED CT reference sets & key concepts

STANDARD CONTAINER

What is a Refset?

A **Reference Set (Refset)** is a standard SNOMED CT release format for customized components, configurations, and metadata.

Flexibly represents subsets, mappings, language preferences, and hierarchies alongside core content.

COMMON DISTINCTIONS

Subsets vs. Value Sets

Subset: A specific set of SNOMED CT components selected for a particular domain or use case (e.g., pick-lists).

Value Set: A broader terminology term for a set of codes used in a specific clinical context (can span multiple code systems).

CROSS-SYSTEM ALIGNMENT

Maps & Associations

Map Refset: Links SNOMED CT concepts to/from external classification systems like ICD-10 or local code lists.

Association Refset: Maintains historical relationships, linking inactive concepts to active replacement terms.

KEY TAKEAWAY: REFSET AS AN OVERARCHING FRAMEWORK

While terms like **subset**, **value set**, and **map** refer to the **functional purpose** of selected content, a **Reference Set** is the uniform, standardized **technical structure** defined by SNOMED CT to represent and distribute all of these derivatives reliably.

how editions and extensions fit together

What an extension is



Concepts, descriptions, relationships and reference sets added on top of the International Edition for national or organisational needs.

Namespace identifiers



Every extension has a namespace issued by SNOMED International, so each component identifier is globally unique and traceable to its issuer.

Dependency



An extension always depends on the International Edition. Combined, they form an edition, the package your terminology server loads.

Versioning



A national edition declares which International Edition version it depends on. Snowstorm resolves RF2 package dependencies on import.

DEPENDENCY STRUCTURE

International Edition

↑ DEPENDS ON

National extension (namespace 1000206)

Together: a national edition.

NAMESPACES IN IDENTIFIERS

22298006

International Concept

no namespace required

201000000106

National Concept

namespace: 1000000

extensions are **not** editions

Modular architecture allows customization while ensuring global interoperability.

BASE CORE

International Edition

The foundational baseline maintained globally.

- Global concept definitions
- Core reference sets
- Universal release schedule

+

ADD-ON MODULES

Extension Modules

Custom local additions bound to modules.

- National / local content
- Tracked by **moduleId**
- Maintained independently

=

DEPLOYABLE UNIT

SNOMED CT Edition

Complete standalone combination.

- International Edition + Extensions
- Complete working dataset
- Ready for production use

KEY TAKEAWAY: All extensions depend on the International Edition. Publishing extension modules together with their base dependencies forms a complete, standalone SNOMED CT Edition.

PART 1 · SECTION 03

terminology server setup

Guided setup of Snowstorm Lite in Docker, with release files retrieved through your MLDS account.



flat lists, or the SNOMED CT structures

INPUT: FLAT LISTS (JSON, CSV)

Codes and display terms only

Single-language input ✓ **supported**

Multi-language input ✗ **not supported**

Decision support ✗ **not supported**

Reporting ✗ **not supported**

Data analytics ✗ **not supported**

Lookup and validation work, but nothing can be inferred, every new requirement means a new list.

INPUT: RF2 RELEASE FILES

The full knowledge graph

Single-language input ✓ **supported**

Multi-language input ✓ **supported**

Decision support ✓ **supported**

Reporting ✓ **supported**

Data analytics ✓ **supported**

Hierarchy, synonyms, languages and historical associations are all available, so value sets can be defined by rule rather than by hand.

what is a terminology server?

CORE ROLE

Centralized Engine

Specialized REST API service for hosting and querying clinical terminologies.

- Exposes FHIR APIs ([\\$lookup](#), [\\$translate](#), [\\$subsumes](#))
- Decouples terminology from core EHRs
- Real-time validation & concept resolution

SNOMED COMPLEXITY

Graph Traversal

Executes complex logic across SNOMED CT's extensive polyhierarchy.

- Navigates millions of dynamic relationships
- Evaluates ECL queries on the fly
- Instant multi-axial subsumption testing

LOCALIZATION & RF2

Editions & Refsets

Manages dependencies across editions, extensions, and local refsets.

- Automates RF2 package imports & versioning
- Resolves module dependencies seamlessly
- Powers localized search & language refsets

WHY IT MATTERS: Relational databases cannot natively handle polyhierarchies or ECL queries. Terminology servers encapsulate this logic to enable scalable, real-time clinical decision support.

why use terminology services?



Effective data entry

Search algorithms are the key to efficient, accurate data capture inside clinical workflows.



Optimised retrieval

Navigating and retrieving from a large polyhierarchy needs indexing built for the job.



Standardised queries

ECL and FHIR operations mean the same query works against a different server tomorrow.



Frequent updates

Editions are versioned and reloaded, so bindings can be revalidated against each release.

Snowstorm and Snowstorm Lite

Snowstorm

`github.com/IHTSDO/snowstorm`

Key features

FHIR terminology services · full SNOMED ECL support · multiple code systems (LOINC, ICD) · scalable via Elasticsearch

Powers

Authoring Platform · SNOMED CT Browser · Analytics Demonstrator · UI demos

Snowstorm Lite (today's server)

`github.com/IHTSDO/snowstorm-lite`

Key features

FHIR terminology services · ECL core query support · multiple code systems · very fast, low memory · simple cross-platform install

Powers

Bahmni EHR (search, CDS, reports) · Analytics Demonstrator · UI demos · SNOMED Mini Browser

Both are distributed under the Apache 2.0 licence.

what Snowstorm is not

No commercial support

SNOMED International is not able to provide a commercial support agreement for Snowstorm or Snowstorm Lite.

No hosted production service

SNOMED International is not able to host an international cloud service for use in production systems.

PUBLIC INSTANCE USE CASE

The public instance is made available exclusively for reference and manual testing. Anything you build for real runs on your own infrastructure, which is why we set one up today.

standards prevent lock-in



Because the interface is a standard, a provider can change server without changing every clinical system that depends on it.



RF2: what you load into the server

Release Format 2

Every release is a zip archive of tab-separated value files in UTF-8.

Snapshot directory

The current state of every component in that edition, what you normally import.

Full directory

The complete version history of every component. Not usually needed.

Edition or extension package

An **edition** package contains everything and has no dependencies (often over 500 MB). An **extension** package depends on the International Edition.

THE FILES THAT MATTER

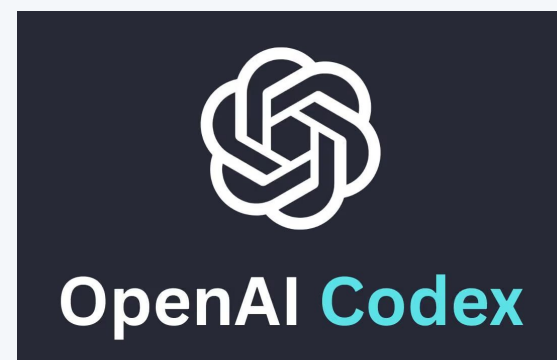
Concept	Master record of every concept, id, active, definition status
Description	FSNs and synonyms, the human-readable terms for the interface
Relationship	The logical model, enables subsumption and ECL
Language refset	Preferred or acceptable per language and dialect
Map refset	SNOMED CT to ICD-10/11, MedDRA or custom targets
Simple refset	Subsets and value sets; module dependencies

Snowstorm Lite setup with an AI Assistant

Use your AI Assistant to guide you through the installation of Snowstorm Lite:

<https://github.com/alopez/snowstorm-lite-guide>

- Will check the system requirements
- Will offer different setup options
- Can install and run anything you need, and drop you in the Snowstorm Dashboard to finish
- Can also work in as an off-hand guide, you type and send all the commands
- Can help you troubleshoot and respond any implementation question



Public

Pin
Watch 0
Fork 0
Star 0

main
3 Branches
0 Tags
Go to file
Add file
Code

Ask the user how hands-on the agent should be (4 levels)
4911792 · 6 hours ago
11 Commits

.gitignore	english branch: general English guide (edition-agnostic)	last week
AGENTS.md	Ask the user how hands-on the agent should be (4 levels)	6 hours ago
LICENSE	LICENSE: copyright a SNOMED International	last week
README.md	Ask the user how hands-on the agent should be (4 levels)	6 hours ago
docker-compose.yml	english branch: general English guide (edition-agnostic)	last week
snowstorm-lite.env.example	english branch: general English guide (edition-agnostic)	last week

README
Apache-2.0 license

Install guide — Snowstorm Lite

Got an AI agent? Start here. Copy this and hand it to your agent (Claude Code, etc.). It will read the instructions, **ask how hands-on you want it** (do everything · set it up and hand off at the dashboard · just guide you while you type · only troubleshoot) and what it needs (your OS, whether you have MLDS credentials, files, or nothing), then guide you:

Help me install Snowstorm Lite step by step.
Clone/read this repo, follow AGENTS.md, run the preflight checks and ask me what you need (OS, MLDS credentials / files / nothing):
<https://github.com/alopez/snowstorm-lite-guide>

About

Guía + Docker Compose para correr Snowstorm Lite (edición español) sin fricción: fix de permisos, AGENTS.md para asistentes de IA, y 3 formas de cargar terminología.

Readme
 Apache-2.0 license
 Activity
 0 stars
 0 watching
 0 forks

Releases

No releases published
[Create a new release](#)

Packages

EXERCISE · HANDS ON

Snowstorm Lite setup

- 01 Start Docker Desktop.
- 02 Pull the Snowstorm Lite image on the command line.
- 03 Create a folder on your desktop named **snowstorm-lite**.
- 04 In a text editor, create an environment file **snowstorm-lite.env** holding your MLDS syndication credentials, and save it in that folder.
- 05 Navigate to the folder and start Snowstorm Lite.

Allow the container at least 1 GB of memory. Your MLDS account supplies the release files; an affiliate licence is required to access SNOMED CT content.

Pre-installed Demo Server:

<https://implementation-demo.snomedtools.org/snowstorm-lite>

```
docker pull snomedinternational/snowstorm-lite:latest
```

```
SYNDICATION_USERNAME=xxx
SYNDICATION_PASSWORD=xxx
ADMIN_PASSWORD=
```

```
docker run -p 8080:8080 --name=snowstorm-lite \
  --env-file snowstorm-lite.env \
  -v snowstorm-lite-volume:/app/lucene-index \
  snomedinternational/snowstorm-lite:latest \
  --index.path=lucene-index/data
```

install **SNOMED CT** into your server

- 01 Check that Snowstorm Lite is running in Docker.
- 02 Open the Snowstorm Dashboard:
enter **localhost:8080** in your browser.
- 03 Use the **Syndication** tab
find and select your national SNOMED CT edition.
- 04 Install it, including the **International Patient Summary** reference set.

WHILE THE IMPORT RUNS

Import takes a few minutes. Use the time to check your MLDS licence is current, and keep the dashboard open, we walk through its screens next.

If the download fails, the cause is almost always credentials in the environment file or a container memory limit below 1 GB.

GUIDED WALKTHROUGH

Snowstorm Dashboard walkthrough

Open the dashboard on **localhost:8080** and follow along through the screens: syndication and imported editions, the value set list, ConceptMaps, and the query tab you will use for ECL.

Everything you upload in the afternoon exercises appears here.

PART 1 · SECTION 04 · 14:25–15:05

expression constraint language

Writing and refining ECL against your own server to define value sets that stay correct across releases.

querying SNOMED CT effectively

SNOMED CT is a semantic network, to derive the clinical concepts you need, you must navigate it.

THE CHALLENGE

360,000+ active concepts connected by *is a* and attribute relationships.

"All fractures of the femur" spans many descendant concepts.

Enumerating members by hand is error-prone and impossible to maintain.

Bespoke traversal code is complex, non-portable and hard to audit.

ECL: THE SOLUTION

A formal syntax for expressing concept queries.

Traverse a hierarchy in one token: `<`, `<<...`

Filter by attribute values with a single refinement.

Combine criteria with Boolean logic: AND, OR, MINUS.

DEFINITION

what ECL is

A formal syntax for representing expression constraints: computable rules that define a bounded set of clinical meanings.

The set is defined by the logic of SNOMED CT, not by an enumerated list of identifiers. When the terminology is updated, the set follows.

```
<< 125605004 |Fracture of bone|
```

Reads as: every concept that is a fracture of bone, together with all of its subtypes. Today, that is several thousand concepts; you never have to list them.

why a query language for SNOMED CT?

EXTENSIONAL



A list of codes

Maintained by hand

Requires manual curation and continuous human intervention for every update.

Silently goes out of date

Becomes stale at every new release without explicit notifications or tracking.

No clinical rationale

Gives no account of why a concept belongs to the set or its structural inclusion rules.

INTENSIONAL



A rule over the terminology

Written once

Define the rule syntax once and re-use it seamlessly across operations.

Automatically re-evaluated

Dynamically updates membership against each new release of SNOMED CT.

Documents clinical intent

The rule itself self-documents the semantic and clinical criteria of the set.

key use cases

01

EHR value set binding

Restrict the valid values for a data element at the point of care, only diabetes subtypes, say, without maintaining a manual list.

02

Subset definitions

Define subsets intensionally, by rule rather than by enumerating every member, so they stay maintainable.

03

Clinical queries and cohorts

Write machine-processable queries that identify all matching expressions across an edition, cohort finding and quality reporting.

04

Concept model constraints

Constrain the range of an attribute in the concept model, which anatomy concepts may fill a finding site role.

building blocks

```
<< 404684003 |Clinical finding (finding)|
```

HIERARCHICAL SCOPE OPERATORS

<<	Descendant or self
<	Descendants only
>>	Ancestor or self
>	Ancestors only
<!	Children
!>	Parents
*	Any concept
^	Member of a reference set

THE FOCUS CONCEPT

<< [**focus**] selects the designated concept **and** every concept below it in the hierarchy.

Because the hierarchy is maintained for you, a new subtype released next month is inside the value set without any edit on your side.

simple constraints

- 01 Retrieve every kind of fracture of bone, including the concept 125605004 |Fracture of bone| itself. How many results?

- 02 Retrieve only the immediate children of 73211009 |Diabetes mellitus|.

- 03 Retrieve all supertypes of 44054006 |Type 2 diabetes mellitus|, excluding the concept itself.

- 04 Retrieve the members of the 131000274102 |Allergy intolerance substance reference set (foundation metadata concept)|. (German Edition)

simple constraints

01 << 125605004 |Fracture of bone|

02 <! 73211009 |Diabetes mellitus|

03 > 44054006 |Type 2 diabetes mellitus|

04 ^ 723264001 |Lateralisable body structure reference set|

refining a query by attribute

```
<< 404684003 |Clinical finding| :  
    363698007 |Finding site| = << 39057004 |Pulmonary valve structure|
```

Attribute filtering :

Refine with attribute–value pairs after a colon.
Format: **focus : attribute = value** . The value can itself be a sub-expression.

Boolean logic

AND or a comma for conjunction, **OR** for disjunction, **MINUS** for exclusion.

Cardinality and groups

[m..n] matches a number of attribute occurrences; **{ }** groups attributes into role groups; **ANY** matches any attribute concept.

other features

01

Member filters

Query reference set membership directly with ^.

```
^ 816080008 |International patient summary|
```

02

Description filters

Filter by term text, language, description type or acceptability.

```
< 404684003 {{ term = "heart attack", type = syn }}
```

03

Concrete values

Constrain attributes with numeric, string or boolean values, not only concept references.

```
< 27658006 : 1142135004 = #400
```

04

History supplements

Include retired concepts linked via historical association refsets, essential for longitudinal reporting.

```
< 404684003 {{ +HISTORY }}
```

executing ECL on a terminology server

ECL EXPRESSION

```
<< 254837009 |Malignant tumor of breast|
```

SERVER EXAMPLES

Snowstorm, SNOMED International

Ontoserver, CSIRO

Snowray, B2i Healthcare, an IQVIA business

EXPANSION

```
254837009 448435005 254840009 286896005  
278050001 271467005 403458008 373082000 ...
```



The same expression, evaluated against a declared **substrate** : a SNOMED CT edition at a specific version.

the same expression can return **different results**

EDITIONS AND EXTENSIONS

-  **International Edition**, the baseline every edition shares.
-  **National extensions**, country-specific additions not present internationally.
-  **Local extensions**, organisation-specific concepts layered on a national edition.

VERSIONS AND RELEASES

-  **Versioned releases**, a version URI uniquely identifies a release.
-  **Concept lifecycle**, concepts are added, inactivated or split, so a result set changes over time.
-  **Reproducibility**, always pin the version URI in production for auditable results.

ECL EXPRESSION EXAMPLE

<< 64572001 |Disease|

INT Jan 26

91.6k

matching concepts

US Mar 26

94.7k

matching concepts

BE May 26

92.4k

matching concepts

one constraint, several substrates

Use the same expression throughout: `< 36456004 |Mental state finding (finding)|.`

-
- 01 Run it against the International Edition, latest release. How many concepts?
 - 02 Switch the edition to the Belgian edition and run it again. Has the count changed? Have the terms changed?
 - 03 Stay on the Belgian edition and select an earlier release. What changes now, and why?
 - 04 Reflecting on your four results: What would you quote in a specification (the constraint, the substrate, the expansion the number of members), and what context is essential alongside it?

one constraint, several substrates

-
- 01 Your baseline expansion. Record the edition and the release date next to it, or the number means nothing to the next reader.
 - 02 The count is the same or slightly higher: a national edition contains all International content, and may add extension concepts. The terms change, because German descriptions are distributed in the extension.
 - 03 The count moves again, this time because of the version rather than the edition. Concepts have been added, inactivated or remodelled between the two releases.
 - 04 Quote the constraint, not the count. If a count is required, state the substrate with it: edition, release date, and the date you ran it.
-

A result without its substrate is not reproducible. This is why ECL expressions, not concept lists, belong in specifications.

15:05–15:20 CEST

break

Reconvening in 15 minutes, Part 2 begins with terminology bindings.

PART 2 · SECTION 05 · 15:20–15:55

terminology bindings

Binding value sets to data elements in an information model, and the FHIR terminology operations that support it.

The Two Pillars of Semantic Interoperability

Terminology binding bridges **structure** and **meaning** to enable quality health information exchange.



Information Model

Defines the **structure and relationships** of clinical data.



Terminology

Provides **systematized vocabulary** with precisely defined meanings.

Pillar 1 - THE INFORMATION MODEL

Defines the structure of health data - which fields exist, what types they hold, and how they relate.

What does it do?

- Structures clinical data into named fields with defined types
 - Defines relationships between data objects (references)
 - Specifies cardinality, required vs optional, etc.
 - Assigns a data type to each field, e.g CodeableConcept
- Standards: FHIR, openEHR, HL7 v2*

FHIR: Condition Resource

● clinicalStatus	Coded
● verificationStatus	Coded
● code	Coded
● subject	Reference

● required ☐ = coded (→ binding)

Key insights:

The model answers:
 "What field is this?"

But NOT:
 "What coded value belongs here?"

This is answered by terminology binding

→ Needs Terminology Binding

HL7 FHIR Information Model Examples

Resources define the specific structure for clinical entities, ensuring consistent data exchange.

FHIR: Patient Resource

Identifies the individual receiving healthcare services.

- `identifier`
- `name [1..*]`
- `gender`
- `birthDate`
- `address`

FHIR: Condition Resource

Records detailed information about a problem or diagnosis.

- `clinicalStatus`
- `verificationStatus`
- `code (SCTID)`
- `subject (Ref)`
- `onsetDateTime`

FHIR: Procedure Resource

Describes actions performed on or with a patient.

- `status`
- `category`
- `code (SCTID)`
- `subject`
- `performed[x]`

Pillar 2 - TERMINOLOGY

Provides systematized vocabulary with precisely defined meanings via SNOMED CT.

What does it do?

- Provides the **semantic meaning** for clinical data elements
- Enables **computable knowledge** through a formal logic-based framework
- Supports clinical decision support, analytics, and cross-border exchange
- Standardizes how clinicians record findings, diseases, and procedures

Standards: SNOMED CT, LOINC, ICD-10/11

SNOMED CT: The Global Language

350,000+ Concepts

Unique clinical meanings (e.g., Myocardial infarction)

1,000,000+ Descriptions

Human-readable terms and synonyms in multiple languages

Relationships (is-a / attribute)

Defining the hierarchy and clinical attributes

Pillar 2 answers: "What does this code mean?"

Terminology in FHIR Model

Coded elements (*CodeableConcept*) within resources provide the "meaning" of the data through terminology binding.

Condition Resource

Captures clinical state, diagnosis, or problem.

clinicalStatus Required

verificationStatus Required

code EXTENSIBLE
<< 404684003 |Clinical finding (finding)|

bodySite Preferred

Procedure Resource

Records action performed on or with a patient.

status Required

category Example

code PREFERRED
<< 71388002 |Procedure (procedure)|

reasonCode Example

FHIR Binding Strengths

How strictly must the value set be followed? FHIR defines four conformance levels.

REQUIRED

MUST use

Implementations **MUST** use a code from the ValueSet. No alternatives.

Used for administrative status codes.

AllergyIntolerance.clinicalStatus

→ required binding to
AllergyIntoleranceClinicalStatusCodes

EXTENSIBLE

SHOULD use

Use a value set code if one exists. Only use an alternative if no appropriate code is in the value set.

Most SNOMED CT bindings.

Condition.code

→ extensible binding to << 404684003
|Clinical finding|

PREFERRED

ENCOURAGED

Implementers are encouraged to use codes from this ValueSet.

Alternatives allowed. Signals community consensus.

Procedure.code

→ preferred binding to << 71388002
|Procedure|

EXAMPLE

ILLUSTRATIVE

The ValueSet is illustrative only. No conformance requirement.

Use during early profile design.

Observation.code

→ example binding to
LOINC/SNOMED entities

Why Does Binding Matter?

Without binding, a FHIR Condition.code field could legally contain any of these — all valid SNOMED CT:

Procedure	113091000	Magnetic resonance imaging (procedure)	× Wrong domain
Body structure	85050009	Bone structure of humerus (body structure)	× Wrong domain
Clinical finding	372244006	Malignant melanoma (disorder)	✓ Correct domain
Substance	735029006	Shellfish (substance)	× Wrong domain
Clinical finding	84757009	Epilepsy (disorder)	✓ Correct domain

➡ Terminology binding restricts each field to the correct SNOMED CT hierarchy - and makes the rule machine-enforceable

Value Set vs. Meaning Binding

Value Set Binding

Constrains the **permitted values** for a specific coded data element.

- Defines a subset of codes (Value Set) from a code system.
- Used to populate input fields like "Diagnosis" or "Procedure".
- Ensures data entry falls within a valid clinical domain.
- Enables automated validation at the point of entry.

Meaning Binding

Defines the **semantic meaning** of the information model artifact itself.

- Established link between a model element and a concept.
- Represents the semantics of the data elements or data groups/resources
- Essential for semantic alignment between different systems.
- Provides context for how to interpret the associated data.

Binding a Simple Fracture Model

MY FRACTURE MODEL

Data Element 1: Fracture Type

Defines the nature of the break.

Examples:

- Open Fracture
- Closed Fracture
- Spiral Fracture

Data Element 2: Location

Specifies the anatomical site and laterality.

Examples:

- Humerus
- Femur
- Left vs. Right

Fracture Type

Fracture Location

Binding a Simple Fracture Model

MY FRACTURE MODEL

Data Element 1: Fracture Type

Defines the morphology of the break.

Value Set Binding:

<< 72704001 |Fracture (morphology)|

Meaning Binding:

Associated morphology

Data Element 2: Location

Specifies anatomical site and laterality.

Value Set Binding:

<< 272673000 |Bone structure|

Meaning Binding:

Finding site

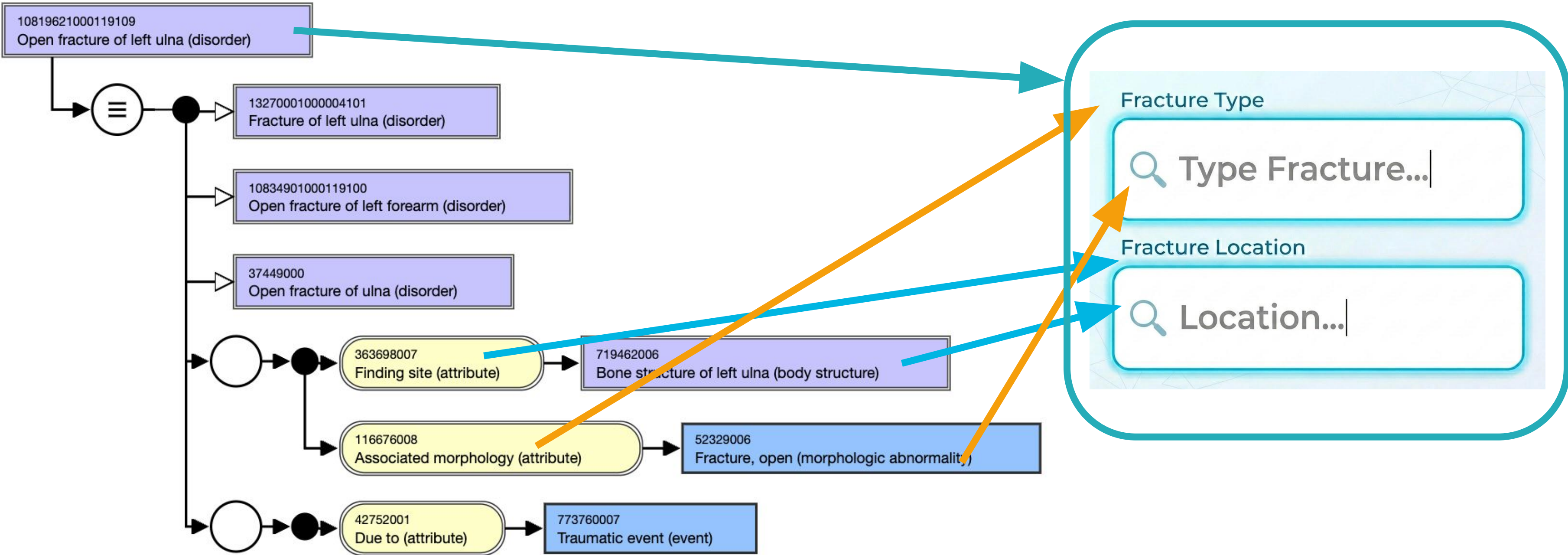
Fracture Type

🔍 Type Fracture...|

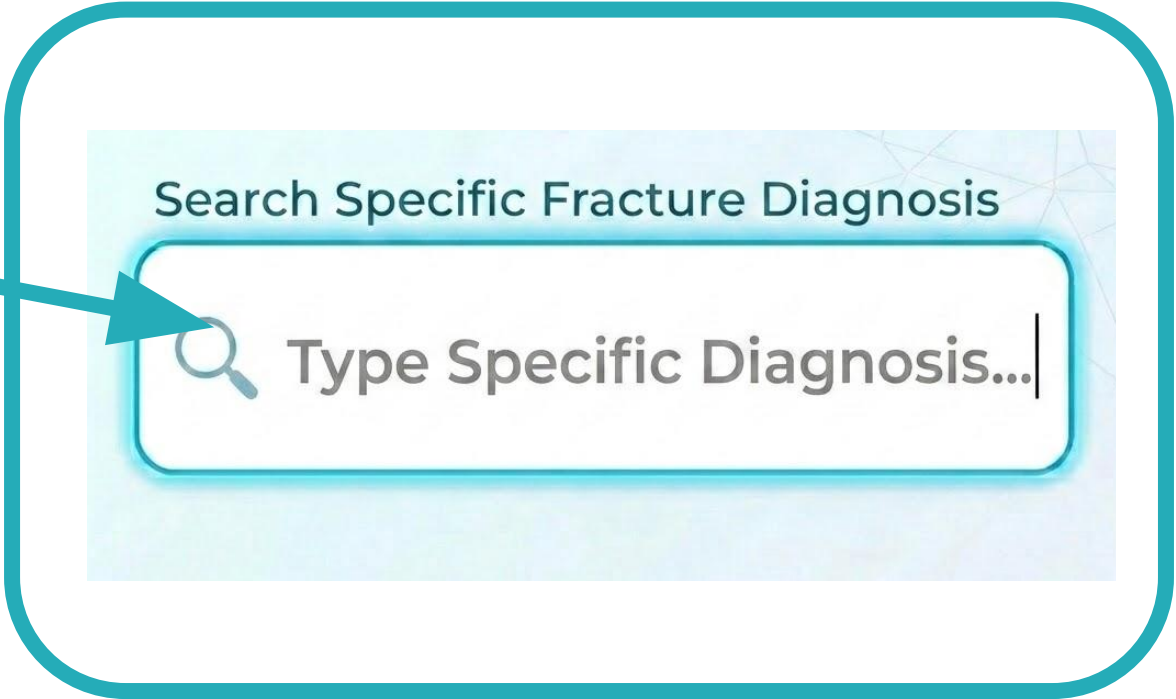
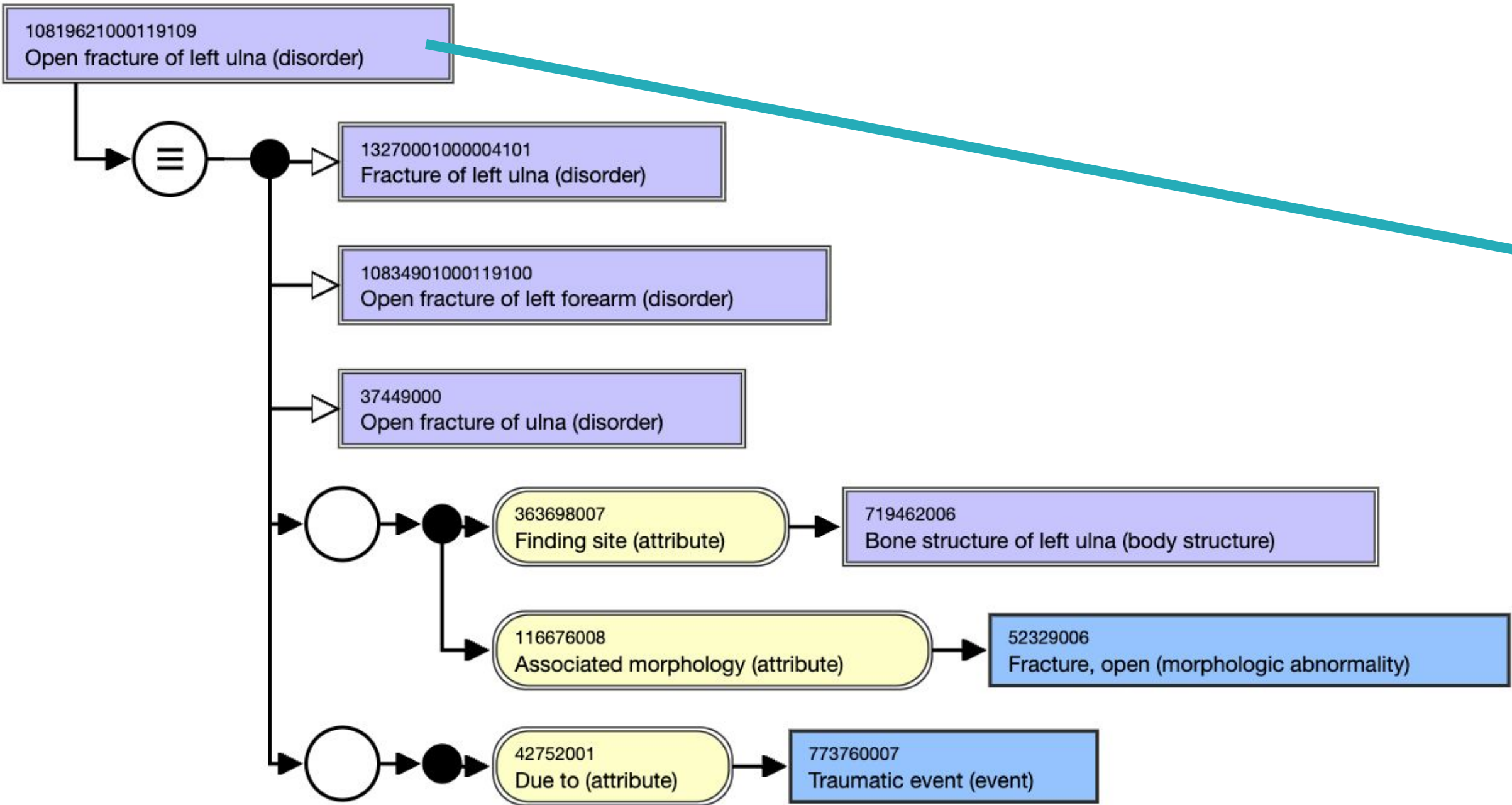
Fracture Location

🔍 Location...|

Binding a Simple Fracture Model



Binding a Simple Fracture Model



< 125605004 |Fracture of bone (disorder)|

Binding Example: FHIR Observation

Meaning Binding

The **Observation.code** element defines the semantic meaning of the observation.

Information Model: Observation

code: CodeableConcept

Maps to SNOMED Attribute:

363787002 |Observable entity|

→ Links the field to the **Observable entity** hierarchy in SNOMED CT.

Value Set Binding

The **Observation.interpretation** represents the result, constrained to a specific set of finding codes.

FHIR ValueSet Binding

strength: preferred

valueSet: "http://snomed.info/sct?fhir_vs=ecl/< 441742003"

Result Domain:

<< **Clinical finding**

→ Restricts allowed result codes to the **Clinical finding** domain.

Meaning binding defines "What is it?" while Value Set binding defines "What are the valid results?"

three ways to define a FHIR ValueSet

ENUMERATED

Explicit list

Codes listed one by one. Highest control, highest effort; members change only by human edit.

```
"concept": [  
  { "code": "113091000" },  
  { "code": "248152002" }  
]
```

Static list · review every release

RULE-BASED

Compose filters

Defined inside the resource with logical filters such as *is-a* or an ECL constraint. Always up to date.

```
"filter": [{ "property":  
  "constraint", "op": "=",  
  "value": "<< 404684003" }]
```

Query based · stored on the server

IMPLICIT

Dynamic URL patterns

No resource exists on the server, a temporary value set is expanded from the request parameters.

```
?fhir_vs=isa/[sctid]  
?fhir_vs=refset/[sctid]  
?fhir_vs=ecl/[query]
```

Query based · runtime only

The FHIR ValueSet Resource

Core metadata elements defining a set of codes from one or more terminologies

url

Canonical identity (URI)

status

active | draft | retired

compose

Content definition

include

System + filter/ECL

expansion

Resolved concept list

```
{
  "resourceType": "ValueSet",
  "url": "http://example.org/fhir/ValueSet/my-valueset",
  "status": "active",
  "compose": { "include": [ { "system": "http://snomed.info/sct" } ] }
}
```

ValueSets decouple the definition of allowed codes from the data elements that use them.

Interpreting Clinical Data

Reliable interpretation requires the union of Structural Context and Semantic Meaning.

1. INFORMATION MODEL

The "Structural Context"

Defines *how* data is organized and *where* it sits within a clinical record.

- **Resource Type:** Condition vs. Procedure
- **Status:** Confirmed, Refuted, Suspected
- **Relationship:** Patient vs. Family Member



2. TERMINOLOGY

The "Semantic Meaning"

Defines *what* the clinical concept is using a standardized clinical vocabulary.

- **Concept Code:** 254837009
- **Fully Specified Name (FSN):** Malignant neoplasm of breast
- **Hierarchy:** Descendants and Ancestors



Conclusion: Awareness of both context and terminology prevents clinical misinterpretation.

one clinical reality, three representations

CURRENT DIAGNOSIS

FHIR Condition

254837009 |Malignant neoplasm of breast|
verificationStatus: **confirmed** clinicalStatus: **active**

FAMILY HISTORY

FHIR FamilyMemberHistory

254837009 |Malignant neoplasm of breast|
relationship: **mother** The model carries the subject context.

SUSPECTED FINDING

FHIR Condition

254837009 |Malignant neoplasm of breast|
verificationStatus: **provisional** Not a SNOMED CT situation concept.

The same code in all three. Context lives in the FHIR model, in status and relationship fields, **not** inside the SNOMED CT concept.

six rules for a durable binding

1 · Know your information model

Cardinality, data type and context determine which hierarchy to bind to.

Do: `Condition.code → << 404684003`

Don't: `Condition.code → << 442083009`

2 · Prefer rule-based ECL

ECL value sets pick up new concepts automatically; static subsets go stale.

Do: `filter: is-a 418038007`

Don't: `enumerate 300 codes by hand`

3 · Choose the strength deliberately

Start with example, move to preferred later. Reserve required for administrative codes.

Do: `extensible for SNOMED CT bindings`

Don't: `required on Condition.code`

4 · Keep context in the model

Do not encode situation context in codes when the model has fields for it.

Do: `Condition + status: provisional`

Don't: `415068001 |Suspected...|`

5 · Avoid grouper nodes

Organisational nodes are not for clinicians. Use leaf or near-leaf concepts.

Do: `<< 64572001 filtered subtree`

Don't: `include 64572001 |Disease| itself`

6 · Match hierarchy to purpose

The hierarchy must fit what the field is for.

Do: `substance → << 105590001`

Don't: `substance → << 71388002`

authoring a binding, step by step

01 Read the data element

Name, data type, cardinality, and what the field is for in the clinical workflow.

02 Choose the hierarchy

Pick the top-level hierarchy that matches the field's purpose, then the narrowest sensible focus concept within it.

03 Write the value set as ECL

Expand it on your server and read the members. Too many, refine with an attribute; too few, widen the focus or drop a filter.

04 Declare the strength

Extensible for most SNOMED CT bindings. Required only where an alternative code would break the receiving system.

05 Test it against real data

Validate codes already in your records. Anything rejected is either a wrong binding or a data quality finding, and both are worth knowing before go-live.

keeping a binding correct across releases

Every SNOMED CT release adds, inactivates and re-parents concepts. A binding that was right in January can be wrong by September without anybody editing it.

Rule-based value sets drift

An ECL value set picks up new matching concepts automatically, which is the point, but it can also silently gain or lose members when the hierarchy changes. Review the expansion after each release.

Enumerated value sets go stale

A fixed list is predictable, but members may be inactivated. Check every code is still active, and add newly relevant concepts deliberately.

Stored data outlives the binding

Codes already in the record may now be inactive. Use historical association reference sets, or the ECL history supplement, when querying across time.

Pin the version, automate the check

Reference a version URI in production for reproducible results, and run the expansion comparison as an automated job rather than a manual review.

PART 2 · SECTION 06 · 15:55–16:30

terminology binding demonstrator

Working through the demonstrator to create, inspect and test bindings against your own terminology server.

open-source examples you can **point at your own server**

EXPLORE THE SHOWCASE

Value Set Utility, Dashboard, translate a list or ECL into a ValueSet.

Terminology Binding Sandbox, create and validate bindings.

FHIR questionnaire manager.

Clinical user interface demonstrations.

[ihtsdo.github.io/sct-implementation-demonstrator](https://github.io/sct-implementation-demonstrator)

WHY THEY HELP

Free to explore, publicly accessible, no licence fees.

Easy to replicate, use them in your own context, connected to your own server.

Built for implementers, targeted at developers doing exactly this work.

connect the tools to *your* server

- 01 Open the SNOMED demonstrators site.
- 02 Switch the FHIR terminology server to **Local server / Custom**.
- 03 Open the EHR lab demo.
- 04 Inspect the transactions, watch **\$expand** and **\$validate-code** being called against your instance.

Demonstrators

<https://ihtsdo.github.io/sct-implementation-demonstrator>

Terminology server

<http://localhost:8080/fhir>

add value sets to *your* server

01 · PREPARATION

Identify two value sets

From a list: export imaging procedures,
tinyurl.com/ValueSets2026

From an ECL:

```
< 442083009 |Anatomical or  
acquired body structure|
```

02 · GENERATE

ValueSet Translator

Open the ValueSet Translator utility.
Upload your exported value set file.
Run the translation.
Export the resulting FHIR ValueSet.

03 · IMPORT AND VERIFY

Snowstorm Lite

Go to the **ValueSet** tab in the dashboard.
Upload the generated files, or generate here.
Confirm both value sets appear in the server's list.

use your value sets in bindings

- 01 Go to the **Terminology Binding Sandbox**.
- 02 Connect it to your terminology server.
- 03 Create a binding: enter a relevant question text and its question binding.
- 04 Set the answer type to **Options**, choose **ValueSet** , and select the value set you uploaded.
- 05 Click **Add binding** and validate the output.

WHAT TO LOOK FOR

The question binding records the **meaning** of the data element.

The answer binding records the **permitted values**.

Validation calls **\$validate-code** against your own server, so the rule is enforceable, not documentation.

PART 2 · SECTION 07 · 16:30–16:50

FHIR ConceptMaps in practice

Building and testing ConceptMap resources to translate local codes to the bound SNOMED CT value sets.

translating **local codes** into bound value sets

What a map does

Links local terms or other code systems to SNOMED CT concepts, so legacy data can be reused without losing its original clinical intent.

How maps are distributed

As map reference sets in RF2, or as FHIR **ConceptMap** resources. Both declare relationships such as equal, broader or narrower.

Direction matters

A map runs either from SNOMED CT to another system, or to SNOMED CT from a legacy terminology. The two are not interchangeable.

THE MAPPING WORKFLOW

- 1 · Extraction**, identify the local terms that need standardising.
- 2 · Tooling**, author in a purpose-built tool such as Snap2Snomed rather than a spreadsheet.
- 3 · Validation**, check the maps against agreed mapping conventions.
- 4 · Deployment**, load into a terminology server for real-time translation.

Maintain a map maintenance strategy from the start: every SNOMED CT release can inactivate a mapped concept.

maps as live resources

REAL-TIME RESOLUTION

Snowstorm hosts maps as active resources, so a clinical system can translate a code through an API call rather than a batch job.

```
GET /fhir/ConceptMap/$translate
?url=http://example.org/fhir/cm/triage
&code=T1&system=http://example.org/local
```

THE CONCEPTMAP RESOURCE

- Metadata**, versioning, status and publisher.
- Elements**, precise source-to-target code pairings.
- Relationships**, equal, broader, narrower.

Because the resource is standard, the same map moves between terminology servers and applications without rework.

Snap2Snomed

Source import with metadata columns, scope control, automapping suggestions, dual-blind authoring with reconciliation, versioning, inactive-concept replacement, and native FHIR ConceptMap export.

snap.snomedtools.org

map exercise: emergency department triage

- 01 Source data** , open the Emergency Department Triage example at tinyurl.com/MappingExercise2026.
- 02 Prepare codes** , make sure terms are unambiguous, and export as CSV or TSV.
- 03 Map** , in Snap2Snomed (snap.snomedtools.org), run the automap, review each row, and export as a FHIR ConceptMap.
- 04 Verify** , upload the file to Snowstorm Lite and confirm the ConceptMap appears in the list, then translate one code.

TRIAGE EXAMPLE DATA

T1	SOB
T2	Abd pain
T3	Fainting
T4	Racing heart
T5	Blood in stool
T6	Throwing up
T7	Prod. cough
T8	Weakness L arm
T9	Speech trouble
T10	Swollen ankle
T11	Feverish
T12	Skin rash

PART 2 · SECTION 08 · 16:50–17:00

wrap-up

Where to take the work next, available support, and follow-up materials.

implementation resources

TERMINOLOGY SERVER

Snowstorm and Snowstorm Lite

RF2 import, FHIR endpoints, ECL queries and language priorities out of the box.

github.com/IHTSDO/snowstorm-lite

EXPLORATION AND QA

SNOMED CT Browser

Browse concepts, hierarchies, descriptions and reference sets across multiple editions.

snomed.org/browser

ELEARNING

Online courses and Training

Self-paced and tutor-led courses, free for Members, providing in-depth knowledge of SNOMED CT features and practices.

snomed.org/courses

DOCUMENTATION

Guides and Specifications

Guidance on translation methodology, ECL, subset creation, mapping and system integration patterns.

docs.snomed.org

THANK YOU FOR ATTENDING

SNOMED
International

questions, and where next

PROVIDE FEEDBACK

Your input shapes these materials and tools.

<https://forms.gle/yFi5HkGXKFEkSGqKA>

OUTSTANDING QUESTIONS

Use the SNOMED forum to reach the
implementation support team.

<https://forums.snomed.org/c/snomed/11>

Reach out to us for follow up
questions:
implementation@snomed.org

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